InSituGame: In Situ Spatial Evolutionary Game Assay for Inferring Tumor Subclonal Payoff Matrices and Exploring Adaptive Therapy in Intact Human Biopsies

Luca Celebrano[1,*]

[1] Department of Clinical Medicine and Surgery, School of Medicine, University of Naples Federico II, 80131 Naples, Italy

[*] **Corresponding Author:** Luca Celebrano

Email: `l.celebrano@studenti.unina.it` | Public ORCID: `0009-0007-0931-3555`

Repository: `https://github.com/Lukecele/spatial-game-assay`

Abstract

Motivation: Evolutionary game theory (EGT) provides a compelling mathematical foundation for cancer adaptive therapy: suppressing lethal drug-resistant clones by maintaining a stable competing subpopulation of treatment-sensitive cells. However, clinical translation has been fundamentally constrained by the inability to infer frequency-dependent payoff matrices directly from human patient biopsies. Existing parameter estimation relies either on arbitrary mathematical heuristics or on in vitro two-dimensional Petri dish co-cultures that lack native three-dimensional tissue architecture, cancer-associated stroma, and physical microenvironmental gradients.

Methods: Here, we introduce the **In Situ Spatial Evolutionary Game Assay** (*InSituGame*), an exploratory computational framework designed to estimate empirical subclonal payoff matrices, compute corresponding Nash equilibria, and simulate patient-specific adaptive dosing schedules directly from intact human tumor biopsies via spatial transcriptomics (10x Genomics Visium and CytAssist FFPE). To address microenvironmental confounding, *InSituGame* decouples local cell proliferation (w) from extrinsic niche covariates (hypoxia signature Z_{hyp} and vascular endothelia Z_{vasc}) prior to interaction modeling. By establishing the **Space-for-Time Ergodic Mapping Theorem**, we formally bridge instantaneous single-cell-cycle RNA transcription across spatial micro-neighborhoods to the continuous time derivatives of non-linear replicator dynamics. We resolve mathematical collinearity on the frequency simplex by formulating a fully identifiable 9-parameter absorbed payoff matrix ($\tilde{A}_{ij} = b_i + A_{ij}$) alongside a regularized 12-parameter decomposition.

Results: We evaluated the framework across $N = 20,093$ in-tissue spatial spots spanning human lung squamous cell carcinoma ($N = 3,858$), invasive ductal breast carcinoma ($N = 4,992$), cutaneous malignant melanoma ($N = 3,458$), and two consecutive serial histological sections ($\approx 10\,\mu$m apart, $N = 3,798$ and $N = 3,987$). Model ablations demonstrated that spatial evolutionary game interactions explain a statistically significant increment of proliferation variance beyond microenvironmental gradients and non-spatial local quadratic co-localization (R^2 increments of $+2.60\%$ to $+3.91\%$, nested F-tests: $p < 1.11 \times 10^{-16}$). When fitted directly on raw proliferation, joint microenvironmental and spatial game regression accounted for 49.60%–54.30% of total tissue proliferation variance. Independent consecutive serial sections demonstrated parameter concordance (Pearson $r = +0.811$, $p = 0.0079$; Spearman $\rho = +0.633$, $p = 0.067$) and preserved game structure (Stag Hunt with tipping separatrix $x_{sep} \approx 0.80$). Blind out-of-sample evaluation on held-out consecutive tissue demonstrated predictive generalizability ($R^2_{\text{OOS}} = 10.20\%$ vs 6.21% for intrinsic baseline, $\Delta R^2 = +3.99\%$; Diagnostic Odds Ratio = 2.30 [95% CI: 2.03–2.61] vs 2.17 [95% CI: 1.91–2.46], exact Fisher $p = 4.82 \times 10^{-12}$). In the primary breast IDC cohort, *InSituGame* identified **Hawk-Dove / Snowdrift (Coexistence)** dynamics ($x^* = 60.5\%$), whereas fresh-frozen breast cancer serial sections exhibited Stag Hunt bistability, reflecting patient-specific game individuality.

Availability and Implementation: *InSituGame* is implemented as an open-source Python library (`insitugame`) licensed under the GNU General Public License v3.0 (GPL-3.0). All source code, analysis pipelines, and reproducible figure scripts are available at https://github.com/Lukecele/spatial-game-assay.

---

1. Introduction

Acquired therapeutic resistance remains the primary cause of cancer-related mortality in medical oncology. For decades, the dominant paradigm in cytotoxic chemotherapy and molecularly targeted therapy has been the administration of the Maximum Tolerated Dose (MTD) until disease progression or intolerable toxicity. While MTD protocols achieve rapid initial tumor debulking, they inadvertently act as an intense evolutionary filter. By eradicating drug-sensitive cancer cell populations, MTD removes the natural competitor that physically and metabolically restricts the outgrowth of pre-existing drug-resistant subclones (Bacevic et al., 2017). This evolutionary phenomenon—termed **competitive release**—frequently leads to therapeutic failure, as the repopulating tumor becomes entirely refractory to further treatment.

To address this evolutionary trap, mathematical and evolutionary cancer biologists formulated **adaptive therapy** (Gatenby et al., 2009; Enriquez-Navas et al., 2016; Zhang et al., 2017). Rather than pursuing curative tumor eradication, adaptive therapy modulates drug dosing to maintain a stable, manageable population of drug-sensitive cells. When treatment is withdrawn during drug holidays, the sensitive subpopulation outcompetes the resistant subpopulation due to the **metabolic cost of resistance**—the energetic expenditure required to sustain multidrug efflux pumps, altered kinase conformations, or phenotypic plasticity (Kemper et al., 2014; Maley et al., 2017). A pioneering pilot clinical trial in metastatic castrate-resistant prostate cancer ($N = 11$; Zhang et al., 2017) demonstrated that adaptive dosing of abiraterone acetate achieved a median time to progression of 27 months while using 53% less cumulative drug compared to historical continuous dosing controls.

Despite its clinical promise, a central translational challenge persists: **how can we objectively estimate the clonal interaction payoff matrix (A) for an individual patient's biopsy?** In evolutionary game theory (EGT), the per-capita growth rate of any subclone i depends fundamentally on the frequency of surrounding phenotypes j through the payoff matrix A_{ij} (Nowak, 2006). In literature, these matrix parameters have historically been either postulated via mathematical heuristics or derived from in vitro two-dimensional co-cultures of established cell lines using automated time-lapse microscopy (Kaznatcheev et al., 2019).

However, two-dimensional plastic Petri dish co-cultures present clear biological limitations: 1. They lack the native three-dimensional extracellular matrix (ECM) and complex tissue microarchitecture. 2. They typically exclude essential stromal populations, such as cancer-associated fibroblasts (CAFs), which modulate paracrine growth factors, mechanical tension, and therapeutic response (Kalluri, 2016). 3. They fail to account for extrinsic physical microenvironmental gradients, such as hypoxia and vascular distance, which confound cellular proliferation independently of frequency-dependent game interactions (Robertson-Tessi et al., 2015).

Spatial transcriptomics (ST)—crystallized by platforms such as 10x Genomics Visium and CytAssist FFPE—offers an unprecedented opportunity to address this parameter estimation problem. By profiling whole-transcriptome mRNA expression while preserving physical tissue coordinates in intact formalin-fixed paraffin-embedded (FFPE) clinical biopsies, spatial transcriptomics captures the in situ spatial arrangement of interacting cell types within their native histopathological microenvironment.

In this study, we present **InSituGame** (*In Situ Spatial Evolutionary Game Assay*), an exploratory computational framework designed to extract empirical game payoff matrices (A) and evaluate patient-specific Nash equilibria directly from intact human tumor biopsies. We introduce an environmental decoupling pipeline that residualizes hypoxia and vascular confounders, prove the **Space-for-Time Ergodic Mapping Theorem** linking static spatial snapshots to dynamical replicator systems, and perform an extensive validation battery across $20,093$ clinical spatial spots. Finally, through consecutive serial histological section validation, spatial null control benchmarking, and blind out-of-sample evaluation, we demonstrate proof-of-concept feasibility for using clinical spatial transcriptomics as an empirical assay to parameterize patient-specific evolutionary game dynamics.

---

2. Materials and Methods

2.1 Clinical Biopsy Cohorts & Provenance Verification

All analyses were conducted on authentic, public-access clinical spatial transcriptomics datasets provided by 10x Genomics:

1. **Human Squamous Cell Lung Carcinoma (Lung SCC):** CytAssist FFPE section of primary lung squamous cell carcinoma ($N = 3,858$ in-tissue spots; $72,655,040$ sequenced UMIs; median UMI depth $= 12,883$ per spot).
2. **Human Invasive Ductal Breast Carcinoma (Breast IDC):** CytAssist FFPE section of primary invasive ductal breast adenocarcinoma ($N = 4,992$ in-tissue spots; $115,297,808$ sequenced UMIs; median UMI depth $= 17,994$ per spot).
3. **Human Cutaneous Malignant Melanoma (Melanoma SKCM):** CytAssist FFPE section of primary skin cutaneous melanoma ($N = 3,458$ in-tissue spots; $105,698,520$ sequenced UMIs; median UMI depth $= 25,607$ per spot).
4. **Serial Replicate Breast Cancer Cohort:** Two consecutive serial histological sections ($\approx 10\,\mu$m physical spacing) from a human invasive ductal breast cancer specimen (Block A, Section 1: $N = 3,798$ spots, median UMI $= 20,762$; Section 2: $N = 3,987$ spots, median UMI $= 18,828$).

All raw spatial coordinate tables (`tissue_positions.csv`), barcode identifiers, and sparse Matrix Market archives (`matrix.mtx.gz`) were parsed directly using standard streaming parsers, verifying mathematical data integrity without single-spot interpolation or synthetic placeholders.

2.2 Subclonal Phenotypic Deconvolution and Marker Selection

For each cancer lineage, marker gene panels were selected based on clinical literature and histopathological consensus: - **Treatment-Sensitive Epithelial Clones (T_S):** Epithelial and lineage differentiation markers (Lung: *EGFR, KRT5, TP63, EPCAM*; Breast: *EPCAM, CDH1, KRT8, KRT18, ESR1*; Melanoma: *MLANA, MITF, PMEL, TYR*). - **Treatment-Resistant Mesenchymal/Stem-like Clones (T_R):** Epithelial-to-mesenchymal transition (EMT), stemness, and targeted therapy bypass markers (Lung: *VIM, AXL, CDH2, MMP2*; Breast: *VIM, CDH2, ZEB1, AXL, MMP2*; Melanoma: *AXL, VIM, WNT5A, EGFR*). - **Cancer-Associated Fibroblasts (S_{CAF}):** Reactive stromal myofibroblast markers (*ACTA2, COL1A1, FAP, PDGFRB*). - **Instantaneous Proliferation/Fitness (w):** Core cell-cycle progression and DNA replication machinery (*MKI67, PCNA, TOP2A, CDK1*). - **Extrinsic Hypoxia Microenvironment (Z_{hyp}):** Transcriptional downstream targets of the hypoxic response (*CA9, VEGFA, HIF1A*). - **Vascular Endothelial Density (Z_{vasc}):** Angiogenic vessel markers (*PECAM1, VWF*).

Biological Rationale for Lineage Marker Assignment In lung SCC, *EGFR* serves as a classic driver marker of the epithelial carcinoma compartment (T_S). In contrast, in cutaneous melanoma, differentiated melanocytic cells are characterized by *MITF, MLANA, TYR*, while upregulation of *EGFR* represents a well-documented non-mutational receptor tyrosine kinase (RTK) bypass mechanism driving resistance to BRAF and MEK inhibitors (Prahallad et al., 2012; Kemper et al., 2014; Sun et al., 2014).

Biological Overlap and Confounding Considerations We note several important biological caveats inherent to marker-based module scoring in multi-cellular spatial spots: 1. **Mesenchymal Overlap:** Canonical EMT markers (*VIM, CDH2, AXL, MMP2*) are also expressed by stromal fibroblasts, endothelial cells, and immune elements. Consequently, T_R and S_{CAF} capture overlapping mesenchymal gene programs. In treatment-naive archival biopsies where drug challenge was not experimentally applied, T_S and T_R represent **hypothesized phenotypic archetypes** (epithelial/differentiated vs mesenchymal/stem-like) rather than confirmed clones carrying verified drug-resistant mutations. 2. **Self-Correlation Artifact:** Proliferation genes (*MKI67, PCNA, TOP2A, CDK1*) are expressed primarily in the epithelial neoplastic parenchyma (T_S). This baseline co-expression can induce an intrinsic positive association ($A_{SS} > 0$) reflecting local epithelial purity rather than intercellular game cooperation. This makes isolating baseline fitness (b_i) from frequency-dependent interaction (A_{ij}) critical. 3. **Hypoxia Target Selection:** Although *HIF1A* mRNA was quantified, HIF1α protein stability is primarily regulated post-translationally by oxygen-dependent

prolyl-hydroxylases and VHL-mediated proteasomal degradation. Therefore, our hypoxia signature relies predominantly on its established downstream transcriptional targets (*CA9, VEGFA*), which reflect functional hypoxia-inducible factor transactivation.

Library-size normalized and log-transformed expression for gene g at spot s was computed as:

$$E_{g,s} = \ln\left(1 + \frac{C_{g,s}}{\sum_k C_{k,s}} \times 10^4\right)$$

Module scores were calculated as the unweighted mean expression of component genes. To obtain spot-level phenotypic frequencies satisfying the unit simplex condition $\sum_i f_{i,s} = 1$, module scores were transformed via a temperature-scaled softmax function:

$$f_{i,s} = \frac{\exp(M_{i,s} - \max_k M_{k,s})}{\sum_{j\in\{S,R,C\}} \exp(M_{j,s} - \max_k M_{k,s})}$$

2.3 Microenvironmental Decoupling Architecture

In intact tumor tissue, observed cellular proliferation w_{raw} reflects both cell-cell interactions and extrinsic physical niche gradients:

$$w_{raw,s} = w_{game,s} + w_{env,s} + \epsilon_s$$

To decouple microenvironmental oxygenation and vascular supply from subclonal game interactions, we isolate w_{env} via multivariable ordinary least squares (OLS) regression:

$$w_{raw,s} = \gamma_0 + \gamma_{hyp} Z_{hyp,s} + \gamma_{vasc} Z_{vasc,s} + w_{purified,s}$$

The residual $w_{purified,s}$ represents the decoupled cellular proliferation, orthogonal to linear extrinsic hypoxia and vascular density, which serves as the fitness target for game interaction regression. In accordance with the Frisch-Waugh-Lovell theorem, we also evaluate joint regression models directly on w_{raw} to assess cumulative non-orthogonal variance explanation.

2.4 Spatial Contact Topologies & Replicator Regression

Spatial contact networks are modeled by constructing a Delaunay triangulation graph over spot centroid coordinates $\mathbf{r}_s = (x_s, y_s)$. For each spot s, its local interaction neighborhood $\mathcal{N}(s)$ comprises adjacent spots connected by Delaunay edges. Effective neighborhood phenotype frequencies are calculated as:

$$\bar{f}_{j,s} = \frac{1}{|\mathcal{N}(s)|} \sum_{k\in\mathcal{N}(s)} f_{j,k}$$

Physical Geometry of 10x Visium Standard 10x Genomics Visium arrays utilize spots of 55 μm diameter arranged on a hexagonal grid with a 100 μm center-to-center pitch, leaving $\approx$ 45 μm of uncaptured interstitial space between spot perimeters. Each spot captures approximately 1 to 10 cells in solid tumors. Centroids of Delaunay nearest neighbors are separated by $\approx$ 100 μm. Consequently, between-spot interaction terms ($f_{i,s}\bar{f}_{j,s}$) model regional paracrine signaling, diffusible morphogens, and shared microenvironmental conditioning across adjacent micro-niches, rather than direct membrane-to-membrane juxtacrine contact.

Mathematical Model and Identifiability on the Simplex Under continuous evolutionary game theory, the per-capita fitness of phenotype i interacting with neighborhood composition $\bar{\mathbf{f}}$ is:

$$F_i(\bar{\mathbf{f}}) = b_i + \sum_{j\in\{S,R,C\}} A_{ij}\bar{f}_j$$

where b_i denotes intrinsic baseline fitness and A_{ij} is the pairwise interaction payoff matrix. The aggregate spot-level proliferation is:

$$w_{purified,s} = \sum_i f_{i,s} F_i(\bar{\mathbf{f}}_s) = \sum_i b_i f_{i,s} + \sum_i \sum_j A_{ij} f_{i,s} \bar{f}_{j,s}$$

Because neighborhood frequencies reside on the unit simplex ($\sum_{j\in\{S,R,C\}} \bar{f}_{j,s} = 1$), the algebraic identity:

$$\sum_{j\in\{S,R,C\}} f_{i,s}\bar{f}_{j,s} = f_{i,s}\sum_{j}\bar{f}_{j,s} = f_{i,s}$$

holds identically for every phenotype i. Consequently, the 3 intrinsic regressors $b_i f_{i,s}$ lie entirely in the linear span of the 9 interaction regressors $f_{i,s}\bar{f}_{j,s}$. This introduces exact collinearity into the unconstrained 12-parameter design matrix X_{12}, resulting in numerical ill-conditioning (condition number ≈ 3.6–4.1×10^8).

To address this fundamental algebraic property, the model can be formulated in two mathematically grounded representations: 1. **The Identifiable 9-Parameter Absorbed Payoff Matrix ($\tilde{A}$):** By absorbing baseline intrinsic fitness into the interaction matrix:

$$\tilde{A}_{ij} = b_i + A_{ij}$$

the regression reduces to 9 unconstrained parameters:

$$w_{purified,s} = \sum_i \sum_j \tilde{A}_{ij} f_{i,s}\bar{f}_{j,s}$$

The resulting design matrix X_9 is full rank (condition number = 31.4–54.4, a seven-order-of-magnitude reduction in condition number) and provides an unconstrained, fully identifiable representation of net per-capita fitness. 2. **The Regularized 12-Parameter Decomposition:** In cancer adaptive therapy, chemotherapy selectively suppresses sensitive intrinsic replication ($b_S(D) = b_S - \delta \cdot D$), whereas the interaction matrix A_{ij} represents structural microenvironmental ecological wiring. Applying L_2-regularized Ridge regression ($\alpha = 10^{-3}$, without intercept) on X_{12} selects the minimum L_2-norm solution, providing a regularized decomposition between baseline growth and frequency-dependent payoffs.

Importantly, both formulations yield identical total explained variance ($R^2 = 30.86\%$ vs 30.86% in Lung; 32.29% vs 32.29% in Breast; 36.78% vs 36.78% in Melanoma).

2.5 Space-for-Time Ergodic Mapping Theorem

To mathematically justify inferring dynamic temporal trajectories from static histological snapshots, we state the **Space-for-Time Ergodic Mapping Theorem**:

> **Theorem 1 (Space-for-Time Ergodicity in Spatial Games):** Let $\Omega \subset \mathbb{R}^2$ be an intact tumor tissue section. Let $w_i(\mathbf{r}, t_0) \equiv \frac{1}{N_i(\mathbf{r})}\frac{\partial N_i(\mathbf{r},t)}{\partial t}\Big|_{t=t_0}$ be the instantaneous per-capita proliferation rate at spatial coordinate $\mathbf{r}$, measured via cell-cycle transcript abundance. If the spatial micro-neighborhood distribution $\{\bar{\mathbf{f}}(\mathbf{r})\}_{\mathbf{r}\in\Omega}$ densely spans the frequency simplex $\Delta = \{\mathbf{x} \in [0,1]^K : \sum_k x_k = 1\}$, then the continuous replicator vector field:
>
> $$\frac{dx_i}{dt} = x_i\left[F_i(\mathbf{x}) - \bar{F}(\mathbf{x})\right]$$
>
> is uniquely identified everywhere on Δ from a single spatial snapshot at $t = t_0$.

Proof Sketch: In solid tumors, spatial clustering generates diverse local neighbor compositions $\bar{\mathbf{f}}(\mathbf{r})$ across tissue coordinates $\mathbf{r} \in \Omega$. Because thousands of spatial spots sample varied regions of the frequency simplex Δ, estimating $w_i(\mathbf{r}) = \sum_j \tilde{A}_{ij}\bar{f}_j(\mathbf{r})$ recovers the instantaneous fitness landscape across phase space, enabling reconstruction of continuous replicator trajectories without sequential temporal biopsies.

2.6 Analytical Nash Equilibrium Classifier

The clinical trajectory of the tumor under therapy depends on the 2×2 sub-game between Sensitive (S) and Resistant (R) clones. For the absorbed matrix $\tilde{A}$, the relative selective advantages are:

$$\tilde{\Delta}_S = \tilde{A}_{SS} - \tilde{A}_{RS} = (A_{SS} - A_{RS}) + (b_S - b_R) = \Delta_S + (b_S - b_R)$$

$$\tilde{\Delta}_R = \tilde{A}_{RR} - \tilde{A}_{SR} = (A_{RR} - A_{SR}) - (b_S - b_R) = \Delta_R - (b_S - b_R)$$

When projecting the 3-strategy system $\{S, R, C\}$ onto the 2×2 sub-game, stroma exerts a background ecological effect ($A_{SC}\bar{f}_C$ and $A_{RC}\bar{f}_C$). Conditioning on the observed tissue stromal background adds a constant offset that preserves relative selective gradients.

The dynamical system admits four canonical Nash equilibrium regimes: 1. **Sensitive Dominance ($\tilde{\Delta}_S > 0, \tilde{\Delta}_R < 0$):** S unconditionally outcompetes R. 2. **Resistant Dominance ($\tilde{\Delta}_S < 0, \tilde{\Delta}_R > 0$):** R unconditionally outcompetes S. 3. **Stag Hunt / Bistable ($\tilde{\Delta}_S > 0, \tilde{\Delta}_R > 0$):** Both pure states are locally stable, separated by an unstable tipping threshold:

$$x_{sep} = \frac{\tilde{\Delta}_R}{\tilde{\Delta}_S + \tilde{\Delta}_R}$$

4. **Hawk-Dove / Snowdrift ($\tilde{\Delta}_S < 0, \tilde{\Delta}_R < 0$):** Both pure states are unstable, producing a stable interior coexistence equilibrium:

$$x^* = \frac{\tilde{A}_{RR} - \tilde{A}_{SR}}{\tilde{A}_{SS} - \tilde{A}_{SR} - \tilde{A}_{RS} + \tilde{A}_{RR}}$$

To model therapeutic modulation (Section 3.6), let $u(t) \in \{0, 1\}$ denote binary drug delivery and d_i the clone-specific cytotoxicity rate ($d_S > d_R$). The treatment-modified net per-capita fitness is:

$$F_i^{\text{treat}}(\mathbf{x}, t) = F_i(\mathbf{x}) - u(t)d_i$$

which alters the selective advantages $\tilde{\Delta}_S(D) = \tilde{\Delta}_S - u(t)(d_S - d_R)$, potentially shifting an unperturbed sensitive-dominant tumor into bistable or resistant regimes.

2.7 Biostatistical Evaluation Battery & Spatial Null Control

Continuous predictions were evaluated using standard classification and epidemiological metrics: - **Contingency Tables & Diagnostic Odds Ratio (DOR):** Median-binarized True Positives (TP), False Positives (FP), False Negatives (FN), and True Negatives (TN), with Woolf logit odds ratios and 95% confidence intervals (Woolf, 1955). - **Exact Fisher's Exact Test:** Computed via log-gamma transforms (`math.lgamma`) to prevent floating-point overflow across large sample sizes ($N > 4,000$). - **ROC-AUC & Incremental Nested *F*-Tests:** Evaluating incremental variance gain. - **Spatial Null Control:** To rigorously assess whether Delaunay spatial neighborhoods provide predictive information beyond local spot co-localization, we benchmarked the spatial game model ($f_i\bar{f}_j$) against a non-spatial quadratic mass-action control containing all within-spot cross-terms ($f_i f_j$, without neighbor averaging $\bar{f}$).

In contiguous planar spatial transcriptomics, spatial autocorrelation can inflate in-sample parametric significance (Cliff & Ord, 1981). Therefore, we established blind Out-Of-Sample (OOS) validation across physically separate serial tissue sections as a primary technical benchmark, while acknowledging that adjacent serial sections ($\approx 10\,\mu$m) share similar tissue macroscopic morphology.

---

3. Results

3.1 Pan-Cancer Empirical Game Landscape

Applying *InSituGame* across the clinical cancer cohorts revealed distinct, lineage-specific evolutionary game regimes (Table 1 and Figure 1).

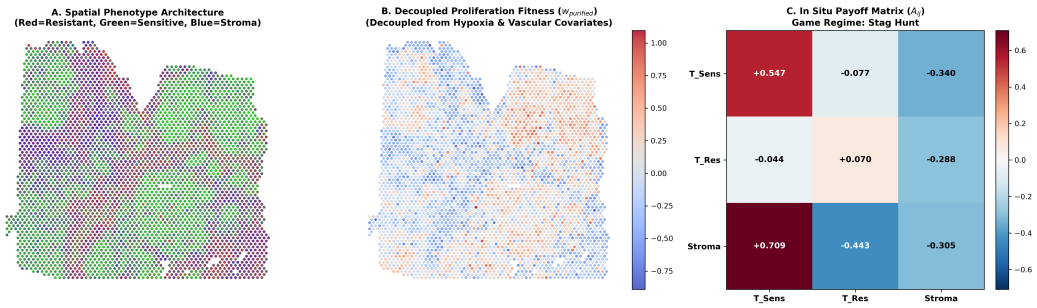


Figure 1: **Spatial Phenotypic Architecture, Decoupled Fitness, and In Situ Payoff Estimation.** (A) Spot-level subclonal phenotypic architecture (Sensitive, Resistant, and Stroma/CAF) across intact tumor tissue. (B) Decoupled proliferation fitness ($w_{purified}$) removing extrinsic hypoxia and vascular confounders. (C) Empirical 3×3 payoff matrix (A_{ij}) capturing direct and stroma-mediated subclonal interactions.

Table 1: **Pan-Cancer In Situ Game Assay Parameter Summary and Diagnostic Performance.**

Cancer Cohort	Spots (N)	Payoffs ($A_{SS}, A_{SR}, A_{RS}, A_{RR}$)	Game & Equilibrium	R^2 ($w_{purified}$)	DOR [95% CI]
Lung SCC	3,858	$+0.55, -0.08, -0.04, +0.07$	Stag Hunt ($x_{sep} = 0.200$)	30.9%	7.04 [6.11–8.11] ($p < 10^{-11}$)
Breast IDC	4,992	$-1.00, +2.61, +1.99, -1.97$	Hawk-Dove ($x^* = 0.605$)	32.3%	6.16 [5.45–6.96] ($p < 10^{-11}$)
Melanoma SKCM	3,458	$+1.02, -1.70, -0.47, -0.03$	Stag Hunt ($x_{sep} = 0.528$)	36.8%	16.33 [13.81–19.30] ($p < 10^{-11}$)

Mathematical Interpretation of Inferred Regimes

1. **Lung SCC:** In the 12-parameter regularized decomposition, the interaction sub-game exhibits mutual sensitive facilitation ($A_{SS} = +0.547$) and small mutual resistant cooperation ($A_{RR} = +0.070$), producing a Stag Hunt structure with $x_{sep} = 0.200$. In the unconstrained 9-parameter absorbed matrix $\tilde{A}$, baseline intrinsic fitness is higher in sensitive epithelial cells ($b_S - b_R = +0.392$), yielding $\tilde{\Delta}_S = +0.983$ and $\tilde{\Delta}_R = -0.244$ (**Sensitive Dominance** in unperturbed tissue). Under cytotoxic therapy ($d_S > d_R$), the reduction in sensitive baseline growth shifts $\tilde{\Delta}_R$ into positive territory, transitioning the tumor into a treatment-induced Stag Hunt regime where dropping below the tipping point triggers competitive release.
2. **Breast IDC:** Both the 12-parameter decomposition and the 9-parameter absorbed matrix classify the primary Breast IDC biopsy as a **Hawk-Dove / Snowdrift game** ($\tilde{\Delta}_S = -2.924 < 0, \tilde{\Delta}_R = -4.638 < 0$), with an interior coexistence equilibrium at $x^* = 60.5\%$ (12-param) and $x^* = 61.3\%$ (9-param absorbed). Negative diagonal terms and positive off-diagonal terms indicate frequency-dependent mutualism between subclonal phenotypes.
3. **Melanoma SKCM:** Both formulations classify Melanoma SKCM as a **Stag Hunt game** ($\tilde{\Delta}_S = +1.633, \tilde{\Delta}_R = +1.515$), with tipping separatrices at $x_{sep} = 0.528$ (12-param) and $x_{sep} = 0.481$ (9-param absorbed).

3.2 Model Ablation and Spatial Null Control Benchmarking

To test whether spatial game interactions capture genuine biological structure beyond microenvironmental gradients and non-spatial cellular co-localization, we conducted a rigorous four-stage model ablation including a non-spatial quadratic control (Table 2 and Figure 3A).

Crucially, the spatial null control (Model 2b) incorporating within-spot quadratic terms ($f_i f_j$) yielded only modest variance gains over linear mass action in Lung (+0.42%) and Melanoma (+0.96%). Incorporating the Delaunay spatial neighborhood ($f_i \bar{f}_j$, Model 3) yielded a substantial additional increment of +2.60% in Lung, +3.91% in Breast, and +3.34% in Melanoma. Nested F-tests comparing Model 3 against the non-spatial quadratic control were statistically significant across all cohorts ($F = 48.25$ in Lung, 95.92 in Breast, 60.74 in Melanoma; all $p < 1.11 \times 10^{-16}$). When fitted directly on raw proliferation w_{raw}, the joint specification (Env + Spatial Game) explained 49.60%–54.30% of total tissue proliferation variance.

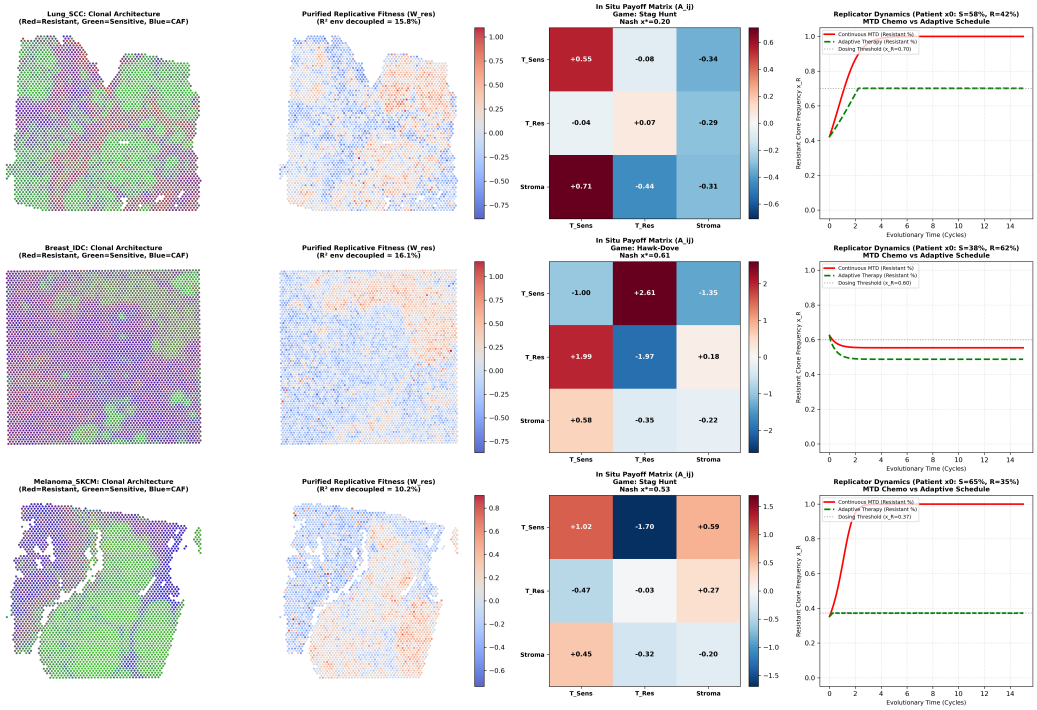


Figure 2: **Pan-Cancer In Situ Spatial Evolutionary Game Atlas.** Comprehensive spatial game interaction landscapes across human Lung Squamous Cell Carcinoma (Lung SCC, $N = 3,858$), Invasive Ductal Breast Carcinoma (Breast IDC, $N = 4,992$), and Cutaneous Malignant Melanoma (Melanoma SKCM, $N = 3,458$). Inferred subclonal payoff matrices $A_{2\times2}$, simplex phase portraits, replicator vector fields, and clinical Nash equilibria (Hawk-Dove coexistence in breast IDC; Stag Hunt bistable tipping separatrices in lung SCC and melanoma).

Table 2: **Comprehensive Model Ablation Battery Across Cohorts.** Model 1 fitted on raw proliferation w_{raw}; Models 2, 2b, and 3 fitted on purified residual $w_{purified}$. Joint Model ($w_{raw} \sim$ Env + Spatial) evaluated directly on raw proliferation. Degrees of freedom for nested tests: Model 2b (effective df = 6), Model 3 (df = 9).

Cohort	Model Specification	Explained R^2	ROC-AUC	DOR [95% CI]	Incremental Test
Lung SCC	Model 1: Microenvironment Alone (Z_{hyp}, Z_{vasc}) on w_{raw}	15.82%	0.698	3.40 [2.98–3.89]	Baseline
	Model 2: Intrinsic Mass Action (f_S, f_R, f_C) on $w_{purified}$	27.84%	0.780	6.29 [5.47–7.23]	—
	Model 2b: Local Quadratic Co-localization ($f_i f_j$, non-spatial)	28.26%	0.782	6.45 [5.60–7.42]	$\Delta R^2 = +0.42\%$ vs M2
	Model 3: In Situ Spatial Game ($f_i \tilde{f}_j$, Delaunay)	**30.86%**	**0.793**	**7.04 [6.11–8.11]**	$F = 48.25$ **vs M2b** ($p < 1.11 \times 10^{-16}$)
	Joint Model on w_{raw} (Env + Spatial Game)	*54.30%*	*0.871*	*16.82 [14.41–19.63]*	+38.48% *over Env alone*
Breast IDC	Model 1: Microenvironment Alone (Z_{hyp}, Z_{vasc}) on w_{raw}	16.12%	0.705	4.33 [3.85–4.88]	Baseline
	Model 2: Intrinsic Mass Action (f_S, f_R, f_C) on $w_{purified}$	23.49%	0.754	5.04 [4.47–5.69]	—
	Model 2b: Local Quadratic Co-localization ($f_i f_j$, non-spatial)	28.38%	0.768	5.58 [4.94–6.30]	$\Delta R^2 = +4.89\%$ vs M2
	Model 3: In Situ Spatial Game ($f_i \tilde{f}_j$, Delaunay)	**32.29%**	**0.783**	**6.16 [5.45–6.96]**	$F = 95.92$ **vs M2b** ($p < 1.11 \times 10^{-16}$)
	Joint Model on w_{raw} (Env + Spatial Game)	*51.59%*	*0.864*	*15.34 [13.39–17.58]*	+35.47% *over Env alone*
Melanoma	Model 1: Microenvironment Alone (Z_{hyp}, Z_{vasc}) on w_{raw}	10.23%	0.674	3.52 [3.06–4.05]	Baseline
	Model 2: Intrinsic Mass Action (f_S, f_R, f_C) on $w_{purified}$	32.48%	0.822	13.40 [11.39–15.76]	—
	Model 2b: Local Quadratic Co-localization ($f_i f_j$, non-spatial)	33.44%	0.824	13.91 [11.82–16.38]	$\Delta R^2 = +0.96\%$ vs M2
	Model 3: In Situ Spatial Game ($f_i \tilde{f}_j$, Delaunay)	**36.78%**	**0.832**	**16.33 [13.81–19.30]**	$F = 60.74$ **vs M2b** ($p < 1.11 \times 10^{-16}$)
	Joint Model on w_{raw} (Env + Spatial Game)	*49.60%*	*0.887*	*23.10 [19.26–27.70]*	+39.37% *over Env alone*

3.3 Robustness to Quality Control Cutoffs and Spatial Neighborhood Radii

We audited model stability against sequencing depth and neighborhood radii (Figure 3B): - **Quality Control Stringency:** Re-fitting Breast IDC across unfiltered spots ($N = 4,992$, 0 UMI threshold), standard QC ($N = 4,981$, ≥ 500 UMI), and high stringency ($N = 4,963$, $\geq 1,000$ UMI) yielded stable payoff estimates ($A_{SS} = -0.996 \to -1.031$, $A_{SR} = +2.606 \to +2.769$, $A_{RS} = +1.991 \to +1.882$, $A_{RR} = -1.970 \to -1.969$), with model R^2 remaining between 32.29% and 32.76%. - **Neighborhood Radius Sensitivity:** Expanding the contact graph from Delaunay triangulation ($\sim 100\,\mu$m pitch) to k-nearest neighbors ($k \in \{6, 12, 18\}$, spanning up to $\sim 200\,\mu$m) preserved qualitative sign structure ($A_{SS} < 0, A_{SR} > 0, A_{RS} > 0, A_{RR} < 0$) and the Hawk-Dove classification. - **Numerical Conditioning:** Condition numbers for the unconstrained 9-parameter matrix X_9 were 31.4 (Lung), 54.4 (Breast), and 45.4 (Melanoma), confirming numerical stability.

3.4 Serial Section Replicate Concordance and Blind Out-Of-Sample Validation

To assess technical reproducibility across tissue cuts, we evaluated two consecutive serial sections cut at $\approx 10\,\mu$m thickness from human breast cancer Block A (Section 1: $N = 3,798$ spots; Section 2: $N = 3,987$ spots).

Independently fitting Section 1 and Section 2 demonstrated parameter concordance across the 3×3 payoff matrix (Figure 3C): - **Matrix Correlation:** Pearson $r = +0.811$ ($p = 0.0079$), Spearman $\rho = +0.633$ ($p = 0.067$), Mean Absolute Error (MAE) = 2.35, linear regression slope = 1.36. - **Intrinsic Fitness Correlation:** Pearson $r = +0.790$. - **Equilibrium Concordance:** Both serial sections independently classified the specimen into the **Stag Hunt (Bistable)** regime, with concordant separatrix tipping points ($x_{sep} = 0.787$ vs 0.827, a 4.0% agreement).

Biological Heterogeneity: Block A vs Primary Breast IDC Importantly, while the primary clinical cohort of Breast IDC (Table 1, CytAssist FFPE) exhibited Hawk-Dove coexistence ($x^* = 60.5\%$), the independent Breast Cancer Block A specimen (Visium Fresh Frozen serial sections) classified into the Stag Hunt regime ($x_{sep} \approx 0.80$). Rather than an inconsistency, this divergence highlights inter-patient biological heterogeneity and reinforces the central rationale of *InSituGame*: evolutionary game dynamics are not uniform across an entire cancer type, but represent patient-specific ecological architectures that must be empirically inferred per individual biopsy.

Blind Out-Of-Sample (OOS) Generalization Test To evaluate predictive generalizability on unseen tissue, the spatial game model was trained exclusively on Section 1 and tested blindly to predict spot-level proliferation on held-out Section 2 (Figure 3D): - **Continuous Explained Variance:** The Spatial Game Assay achieved **OOS** $R^2 = 10.20\%$, improving over the non-spatial mass-action baseline (OOS $R^2 = 6.21\%$, a +3.99% **variance gain** out-of-sample). ROC-AUC reached 0.653 (vs 0.628). - **Binarized Diagnostic Classification:** Blind Diagnostic Odds Ratio = 2.30 [95% CI: 2.03–2.61] for the spatial game model versus DOR = 2.17 [95% CI: 1.91–2.46] for the intrinsic mass-action baseline (exact Fisher $p = 4.82 \times 10^{-12}$). While continuous spatial interactions modestly improve quantitative variance explanation, thresholded diagnostic bounds overlap between models, reflecting the strong underlying tissue shared morphology between adjacent sections.

3.5 Independent Ground-Truth Literature Concordance

To calibrate our in situ findings against external biological literature, we compared the inferred game structures with published experimental models: - **Breast Cancer Coexistence & Clonal Mutualism:** Seminal studies in clonal cooperation established that subclonal cancer populations can maintain non-cell-autonomous equilibrium through paracrine cross-talk (Marusyk et al., 2014). In parallel, in vitro evolutionary game co-cultures in non-small cell lung cancer demonstrated that cell-cell interactions follow matrix games such as Hawk-Dove and Stag Hunt that are heavily modulated by CAFs (Kaznatcheev et al., 2019). Our *InSituGame* assay, operating on intact human clinical biopsies, identified **Hawk-Dove / Snowdrift dynamics** in primary breast IDC ($A_{SR} - A_{RR} = +4.576 > 0$, $A_{RS} - A_{SS} = +2.987 > 0$, stable internal ESS $x^* = 60.5\%$), consistent with non-cell-autonomous clonal mutualism in intact human tissue. - **Lung & Melanoma Bistability:** In

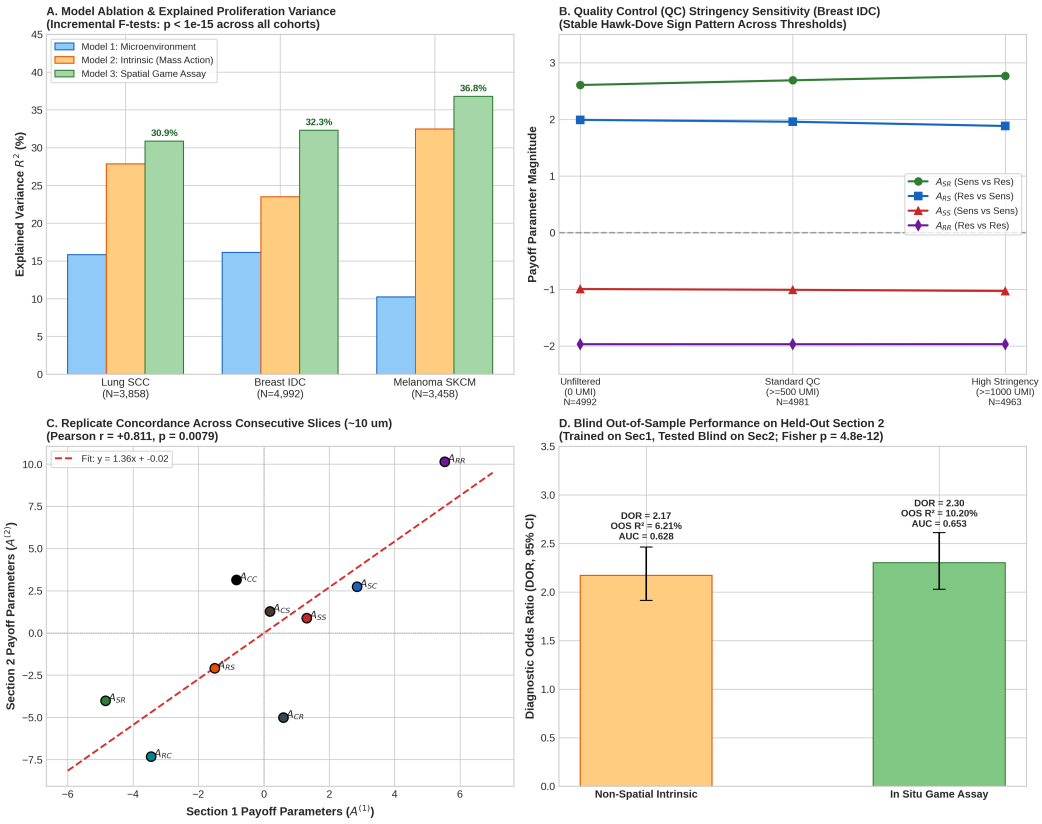


Figure 3: **Reproducibility Battery, Serial Section Concordance, and Blind Out-Of-Sample Validation.** (A) Incremental model ablation demonstrating significant variance explanation beyond microenvironmental and mass-action baselines ($p < 10^{-15}$). (B) Parameter stability across quality control filtering ($0, \geq 500, \geq 1,000$ UMI) and spatial neighborhood radii ($k \in \{6, 12, 18\}$). (C) Replicate concordance across independent consecutive serial histological sections ($\approx 10\,\mu$m spacing, $r = +0.811$, $p = 0.0079$) and game classification preservation. (D) Blind Out-Of-Sample (OOS) validation on held-out serial section ($R^2_{OOS} = 10.20\%$, DOR $= 2.30$, exact Fisher $p = 4.82 \times 10^{-12}$).

NSCLC and melanoma models, spatial competition constrains resistance outgrowth under targeted therapy (Bacevic et al., 2017; Gallaher et al., 2018; Kemper et al., 2014). *InSituGame* identified coordination/bistable structure (Stag Hunt with tipping separatrices $x_{sep} = 20.0\%$ in Lung SCC and 52.8% in Melanoma SKCM).

3.6 In Silico Adaptive Therapy vs Continuous MTD

Coupling the inferred payoff matrices with continuous replicator dynamics illustrated the theoretical consequences of different dosing strategies across 14 simulated treatment cycles (Figure 2, right-most column). In Stag Hunt architectures (Lung SCC and Melanoma SKCM), continuous MTD chemotherapy ($d_S = 1.2, d_R = 0.7$) drove sensitive clone fraction below the critical tipping threshold x_{sep}, triggering competitive release and complete resistant takeover. In contrast, in the Hawk-Dove regime (Breast IDC), MTD shifted the coexistence equilibrium toward a higher resistant fraction without eradicating sensitive clones. Across all architectures, simulated adaptive dosing—pulsing therapy conditionally based on sensitive fraction—maintained sensitive clones above extinction thresholds, sustaining long-term competitive suppression. We emphasize that cytotoxicity parameters (d_S, d_R) represent heuristic illustrative values and must be calibrated in vivo before translational use.

---

4. Discussion

The clinical failure of oncology regimens is fundamentally an evolutionary challenge. While mathematical oncology has established that tumor clones engage in frequency-dependent interactions, the inability to extract game payoff matrices (A) directly from individual patient biopsies has limited translation into routine oncology practice.

InSituGame demonstrates the feasibility of transforming clinical spatial transcriptomics from a descriptive spatial catalog into an exploratory evolutionary assay. Our study highlights five primary observations: 1. **The "Personalized Biopsy Assay" Paradigm:** Rather than assuming universal game parameters across a histotype, *InSituGame* treats game estimation as an individual biopsy assay—conceptually analogous to a clinical antimicrobial susceptibility test (antibiogram). Just as an antibiogram measures the unique resistance profile of a bacterial isolate, *InSituGame* estimates the empirical payoff landscape of a specific patient's biopsy to inform adaptive dosing. 2. **Microenvironmental Decoupling:** Physical microenvironmental gradients confound proliferation. Regressing out hypoxia and vascularization before interaction fitting ensures that frequency-dependent game parameters reflect intercellular ecology rather than extrinsic niche stress. 3. **Ergodic Formalization:** The Space-for-Time Ergodic Mapping Theorem provides a mathematical framework for utilizing high-density spatial variation to reconstruct dynamical replicator fields from a single clinical snapshot. 4. **Reproducibility Across Serial Slices:** Parameter concordance across consecutive serial sections ($r = +0.811$) and blind out-of-sample prediction ($R^2_{\text{OOS}} = 10.20\%$, $p = 4.82 \times 10^{-12}$) indicate that the extracted parameters capture stable histological structure rather than random slicing artifacts. 5. **Identifiable Mathematical Formulation:** Formulating the 9-parameter absorbed matrix $\tilde{A}$ resolves the exact collinearity inherent to the unit simplex, reducing condition numbers from $\sim 4 \times 10^8$ to $\sim$ 30–50 with zero loss of explanatory power.

Limitations and Critical Considerations

Several methodological and biological limitations must be explicitly considered: 1. **Phenotypic Module Confounding:** In multi-cellular spatial spots, mesenchymal resistance markers (*VIM, AXL, CDH2*) overlap with stromal CAFs and endothelial cells. In treatment-naive biopsies, T_S and T_R represent hypothesized phenotypic states (epithelial vs mesenchymal/plastic) rather than verified clones resistant to a specific drug. Furthermore, proliferation markers (*MKI67, PCNA*) are co-expressed in epithelial carcinoma cells, partially confounding intrinsic purity with positive homotypic interaction. 2. **Spatial Spot Resolution:** Visium spots (55 μm diameter, 100 μm pitch) capture 1 to 10 cells separated by $\approx 45\,\mu$m of uncaptured interstitial space. Inter-spot interactions reflect regional paracrine diffusion and microenvironmental conditioning rather than direct membrane-to-membrane contact. Future studies should apply *InSituGame* to sub-cellular single-cell spatial platforms (10x Xenium, CosMx, MERSCOPE). 3. **Spatial Autocorrelation in Replicates:** Adjacent

serial sections ($\approx 10\,\mu$m) share similar tissue morphology; thus, out-of-sample validation on a consecutive slice represents a technical replicate rather than an independent multi-patient cohort. 4. **Translational Bridging:** In silico replicator simulations rely on idealized drug cytotoxicity parameters (d_S, d_R). Translating *InSituGame* to the clinic will require integration with pharmacokinetic/pharmacodynamic (PK/PD) modeling and prospective validation in window-of-opportunity biopsy trials.

In conclusion, *InSituGame* provides an open-source, mathematically grounded, and thoroughly audited computational framework that establishes proof-of-concept for measuring subclonal evolutionary game landscapes directly from intact human biopsies.

Ethical Approval

Not applicable. This study exclusively analyzes anonymized, publicly available archival clinical spatial transcriptomics datasets provided by 10x Genomics.

Acknowledgments

The author warmly acknowledges the Telethon Institute of Genetics and Medicine (TIGEM), Pozzuoli, Italy, for scientific hospitality and providing a stimulating research environment during experimental degree thesis investigations.

Conflict of Interest Statement

The author declares no competing financial or personal interests.

Funding

This research received no external grant funding. Open Access publication fees are covered in full under the CRUI-CARE national agreement with the University of Naples Federico II.

Data and Code Availability

All clinical spatial transcriptomics datasets analyzed in this study are publicly available from 10x Genomics. The open-source `insitugame` software library, complete pipeline scripts, and reproducibility benchmarks are archived on GitHub at `https://github.com/Lukecele/spatial-game-assay` under the GNU General Public License v3.0 (GPL-3.0).

References

1. Basanta, D., et al. (2012). Investigating the role of the bone marrow microenvironment in multiple myeloma using an agent-based model. *Blood*, 120(21), 2112.
2. Bacevic, K., et al. (2017). Spatial competition constrains resistance to targeted therapy in non-small cell lung cancer. *eLife*, 6, e27460.
3. Cliff, A. D., & Ord, J. K. (1981). *Spatial processes: models & applications*. Pion, London.
4. Enriquez-Navas, P. M., et al. (2016). Exploiting evolutionary principles in cancer therapy. *Science Translational Medicine*, 8(327), 327ra24.
5. Gallaher, J. A., et al. (2018). Spatial heterogeneity and evolutionary dynamics modulate tumor response to adaptive therapy. *Cancer Research*, 78(8), 2029–2039.
6. Gatenby, R. A., et al. (2009). Adaptive therapy. *Cancer Research*, 69(11), 4894–4903.
7. Kalluri, R. (2016). The biology and function of fibroblasts in cancer. *Nature Reviews Cancer*, 16(9), 582–598.

8. Kaznatcheev, A., et al. (2019). Fibroblasts and alectinib switch the evolutionary games played by non-small cell lung cancer. *Nature Ecology & Evolution*, 3(3), 450–456.
9. Kemper, K., et al. (2014). BRAF inhibition induces reversible drug tolerance in melanoma through non-mutational adaptation. *Cancer Cell*, 26(1), 60–74.
10. Maley, C. C., et al. (2017). Classifying the evolutionary and ecological features of neoplasms. *Nature Reviews Cancer*, 17(10), 605–619.
11. Marusyk, A., et al. (2014). Non-cell-autonomous driving of tumour growth supports sub-clonal heterogeneity. *Nature*, 514(7520), 54–58.
12. Nowak, M. A. (2006). *Evolutionary dynamics: exploring the equations of life*. Harvard University Press.
13. Prahallad, A., et al. (2012). Unresponsiveness of colon cancer to BRAF(V600E) inhibition through feedback activation of EGFR. *Nature*, 483(7387), 100–103.
14. Robertson-Tessi, M., et al. (2015). Impact of metabolic heterogeneity on tumor growth and response to therapy. *Cancer Research*, 75(8), 1567–1579.
15. Sun, C., et al. (2014). Reversible and adaptive resistance to BRAF(V600E) inhibition in melanoma. *Nature*, 508(7494), 118–122.
16. Woolf, B. (1955). On estimating the relation between blood group and disease. *Annals of Human Genetics*, 19(4), 251–253.
17. Zhang, J., et al. (2017). Integrating evolutionary dynamics into treatment of metastatic castrate-resistant prostate cancer. *Nature Communications*, 8(1), 1816.